Crime Deterrence and Surveillance System



2006 Operations Manual

Welcome to the Team!

From all of us at ICx New-Heights Corporation, congratulations on your decision to become a part of the Sky Watch Crime Deterrence and Surveillance System network. We are dedicated to insuring protection and peace of mind for people wherever Sky Watch products are being used. This manual is designed to help you obtain the highest level of satisfaction from your Sky Watch tower. It provides information regarding safety precautions, operations, troubleshooting, plus additional information in case you need to contact us. It is our goal to make sure that you are completely satisfied with your Sky Watch product. Once again we want you to know how pleased we are that you are a part of our efforts to improve quality of life by providing the very best products in crime deterrence and surveillance.
We are glad you are part of our Team!

COMPANY MISSION

To be the premier manufacturer of manned and unmanned surveillance equipment in the areas of product innovation, exceptional service and quality while meeting the company's and its employee's financial goals.

MARKET VALUE PROPOSITION

The ICx New Heights core value proposition is **Customer Intimacy**. This is an approach characterized by strong customer relationships, high customization and strong collaboration with the customer.

NEED SUPPORT?

As part of the ICx New Heights team, you have access to a skilled support staff available to assist you in any situation.

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Customer Support Director
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Section 1: Safety Precautions

Before operating Sky Watch, read this Operations Manual and become familiar with it and your equipment. Safe and efficient operation can be achieved only if the equipment is properly operated and maintained. Many accidents are caused by failure to follow fundamental rules and regulations.

The following symbols are found throughout this manual. These symbols are to alert you to conditions which are potentially harmful or fatal to the operator, service personnel or the equipment. Review each symbol and be familiar with what each one represents.



"DANGER" This symbol refers to immediate and dangerous hazards which can result in severe injury or death.



"WARNING" This symbol refers to a hazardous or unsafe practice which can result in severe personal injury or death.



"CAUTION" This symbol refers to a hazardous or unsafe practice which can result in personal injury, product damage or property damage.

Fuels, electrical equipment, batteries, exhaust gasses and moving parts present potential hazards that could result in serious personal injury. Take care in following these recommended procedures. Be sure to review the hazards found in each of the accessory equipment owners' manuals in addition to this section. These manuals are included.

FUEL AND FUMES
Fire and explosions can result from improper practices.

- Refer to the generator owners' manual for hazards associated with fuel usage.
- DO NOT store open or unsealed fuel containers on the tower while the generator is in operation or the tower is being transported.
- DO NOT smoke or use an open flame around the generator or fuel tank.

EXHAUST GASES
These gases are DEADLY!

- The on-board generator engine exhaust gases contain CARBON MONOXIDE, a gas that can be potentially lethal.
- DO NOT operate Sky Watch in any type of enclosed area, while utilizing the on-board generator. To operate Sky Watch indoors, power is supplied to tower via power cords (Shore Power).

MOVING PARTS
Can cause severe personal injury or death.

- Keep hands away from any moving parts or pinch points.
- Before descending, check to make sure nothing is in the way of the officer compartment descent path.

Section 1: Safety Precautions (cont.)

ELECTRICAL SHOCK
Can cause severe personal injury or death

- Turn off main power switch before attempting to service any of the tower's electrical equipment. Use rubber insulated mats, placed on dry wood platforms over floors that are metal or concrete, when around electrical equipment.
- Do not wear damp clothing (particularly wet shoes) or allow exposed skin to be damp when handling electrical equipment.
- Use extreme caution when working on electrical equipment. High voltages can cause injury or death.
- Follow state and local electrical codes. Have all installations, of electrical connections, performed by qualified licensed electricians. Tag all open switches to avoid accidental closure and shock.

GENERAL SAFETY PRECAUTIONS

- Benzene and lead, found in some gasoline, have been identified by certain state and/or federal agencies as causing cancer or reproductive toxicity. When checking, draining or adding fuel to your Sky Watch, take care not to ingest, breathe the fumes or allow direct skin contact.
- Used engine oils have been identified by some state and/or federal agencies as causing cancer or reproductive toxicity. When checking or changing engine oil, take care not to ingest or allow direct skin contact with used oil.
- Do not work on this equipment when mentally/physically fatigued or after consuming alcohol or drugs that may impair safe operation of the tower.
- Your Sky Watch tower has been designed for security and surveillance purposes ONLY. Do not use the tower for anything that would compromise the safety of personnel or the tower.



Do not attempt to deploy, use or operate any part of this tower until this manual has been read thoroughly.

- This is not an anti-ballistic tower. No claims or assertion of anti-ballistic capability are made or implied. Proper assessment of the threat level of any situation or area must be made prior to deployment. Failure to comply with this directive may result in injury to personnel and damage to the tower for which ICx New Heights Corporation assumes no responsibility.
- ICx New Heights Corporation is not responsible for injury or damage resulting from non-standard equipment installations or modifications added to the tower by anyone other than ICx New Heights Corporations Factory Authorized Personnel.



"DANGER" Do not deploy this tower under power lines, electrical or communications wires. The ascent path of the officer compartment may come in contact with these wires and other obstructions. Make sure the tower is not set up within fifty (50) feet of any electrical lines.



"DANGER" Do not deploy this tower on sloping or uneven terrain.

"DANGER" Do not deploy this tower in water. This tower is designed and intended to be deployed only on a firm surface. If wet or rainy conditions persist, a proper assessment must be made regarding the necessity to relocate the tower.

Section 2: Glossary
(Definition of Terms)



Breakaway Control Devise: Designed to engage the electrical brakes on the trailer, in the event the tower becomes separated from the towing vehicle while in transit. The key, which activates the breakaway brakes, is located on the side of the tongue. When the key is removed the brakes are engaged and will stay engaged until the key is replaced. Failure to replace this key will drain the on-board batteries.



GFI Outlets: These outlets are designed to automatically open the circuit should they become overloaded or grounded in water.



Outriggers: Used to give the unit its stability and are necessary for safe deployment. During deployment, the outriggers are to be locked into the open position for maximum safety and stability.



Safety Chains: A safety measure used during the transport of the tower. These chains keep the trailer connected to the tow vehicle in case the trailer becomes uncoupled during transport.



Split-ball Coupler: Designed to secure the trailer to the tow vehicle.



Outrigger Lock Pins: These pins lock the outriggers into place during deployment and also during transport.



Stabilizer Jacks: Secure the tower and help provide maximum safety and stability when properly deployed. These jacks are at the end of each of the outriggers.



Stabilizer Jack Adjuster Pin: Designed to allow the height of the stabilizer jack to be adjusted quickly.



Tail Light Hookup: Links the electrical system of the tow vehicle to the Sky Watch tower during transport.



Tongue Mounted Jack: Designed to lift the trailer off the tow vehicle and is also used for aiding in leveling the tower for deployment.

Section 3: Tow Vehicle Requirements

This section describes the minimum requirements needed to safely tow the Sky Watch tower that you have purchased from ICx New Heights Corporation. Along with these requirements, you must check with your State and Local authorities to verify that the vehicle and tow equipment meet the requirements.



Tow Vehicle: ¾ ton or heavier vehicle (Pick-up truck or SUV) for standard models only



Must use One ton or larger for Anti- Ballistic models.



Class III or heavier hitch with a 2" tow ball

NOTE: A Class III hitch is a frame mounted receiver style hitch, this is not the same as a bumper hitch.



7 pin receiver plug (for turn signals, running lights, stoplights and electric brakes)



Electric Brake Controller. All Sky Watch towers are equipped with electric brakes. The Electric Brake Controllers are generally sold as an aftermarket item for the tow vehicle.



Outside Rear View Mirrors. Must be wide enough to see the rear corners of the Sky Watch tower. (Approximately 22 feet from the bumper of the tow vehicle)



Safety chain hook receivers (generally incorporated in the hitch)



2006 Frontier

Section 4:
Coupling/Uncoupling Tow Vehicle



Sky Watch is equipped with an easily operated split-ball coupler. This section covers the function and operation of the hitch and tongue-mounted jack. Refer to this section for directions on coupling and uncoupling the tow vehicle. It is recommended that you become familiar with this section before attempting to couple or uncouple Sky Watch. Failure to do so may result in personal injury, death or damage to the unit.

Uncoupling Sky Watch from Tow Vehicle

1. Chock one wheel on each side, using wheel chocks provided.
2. Disconnect the 7 pin trailer plug, remove safety chains and remove breakaway brake cable from tow vehicle.
3. Turn the tongue jack handle (clockwise) until the foot touches the ground.
4. Release the split-ball coupler
    - A. Remove couplers safety pin.
    - B. Grip the retaining sleeve with two hands and pull back (away from the vehicle) until the spring loaded coupler releases.
5. Return to the tongue jack and continue to rotate (clockwise) until the coupler is released from the hitch.
6. Drive tow vehicle away from the Sky Watch tower.
7. Proceed with stabilizing the tower.

Coupling Unit to Tow Vehicle

1. After making sure the officer compartment is closed and locked.
2. Raise the stabilizer jacks to their limit.
3. Raise the stabilizer jack extensions, by pulling the stabilizer Jack adjusting pins out and lifting the extension, then release pin.
4. Using the tongue jack, adjust the height of the tongue for connection to the tow vehicle. Proceed to back the tow vehicle under the split ball coupler.
5. When the hitch is centered below the split ball coupler, stop the vehicle.
6. Rotate the jack handle counter clock wise until the coupler is in contact with the 2” tow ball.
7. Using your foot, press the split ball coupler’s right side until the locking collar deploys around the coupler.
8. Continue to rotate the jack handle counter clock wise until the jack comes to a stop.
9. Place the safety pin in the coupler.
1.
2. Attach the safety chains, using the cross chain method, to the tow vehicle.
3. Attach the 7 pin connector.
4. Check the tail light operation.
5. Remove wheel chocks.

Section 4:
Coupling/Uncoupling Tow Vehicle

15. Test Trailer brakes.
Make sure safety chains and running lights are connected to the towing vehicle before attempting to tow the tower.

Check light hook-up for full functions prior to transport.
When storing any equipment on the trailer, keep it secured and centered in front of the axle for best towing results.

Do not put any additional equipment inside the officer compartment.

Section 5: Set-Up

Do not attempt to elevate Sky Watch until all outrigger and adjustable stabilizers have been properly deployed. Failure to comply may result In personal injury or death.
Make sure the tires are properly chocked before attempting to elevate the tower.

Make sure the tower is deployed on firm and level surfaces. Avoid soft or uneven surfaces.
For maximum stability, the tower is to be used fully elevated or fully lowered during deployment.
When the tower is deployed, the area around the tower should be roped or coned off for added safety.

Stabilizing Procedures

1. Make sure that one wheel is chocked on each side of trailer.
2. Deploy the outriggers to the outer most limits.
   - A. Pull the locking pin up.
   - B. Rotate the outrigger to outer most limits.
   - C. Make sure the locking pin is in place, pin will lock itself into place.
   - D. Pull stabilizer jack adjuster pin' The extension will drop until at Maximum extension or contact with ground. Release extension pin.
   - E. Rotate the stabilizer jack handle clockwise. Continue unit the extension pin Engages the extension.
   - F. Repeat these steps on the remaining outriggers.
3. With the outriggers deployed and in firm contact with the ground, level the tower.
   - A. Using the installed leveling bubbles , level the tower using the stabilizer jacks to place the bubbles in between the lines.
4. The tower is now stabilized.

Preparing for Operation

1. Unlock the technical compartment, open left door.
2. On the left side of the compartment, locate the rotary switch labeled "Battery Master" Rotate to the "ON" position.
   At this time the Sky Watch unit is ready for operation with the Officers Compartment resting on the transportation gate. This is an acceptable location to operate the tower from. Continue the following steps for operation from ground level.
3. On the left side of the technical compartment, locate the orange control pad (see figure 10). Press and hold the "UP" button until the officer compartment is approximately 5 feet Off the ground.
4. Walk to the officer compartment, just past the fender, stop and lift the transportation gate until it is resting on the rubber bumpers of the crossbar.

"WARNING" The transportation gate is heavy; get assistance to prevent injury to personnel.

5. From the technical compartment, on the orange control pad, press and hold the "DOWN" button until the officer compartment has completely rested on the ground. At this time the tower is ready for operation from ground level.

Section 6: Operation

Sky Watch is designed to elevate one officer from ground level to an eye level of approximately twenty four (24) feet (depending on model). This section covers the operation of the tower. It is recommended that the operator be completely familiar with the tower for safe operation.

Pre-Operation Checks

Before operating the tower, check the ascent (up) path of the officer compartment and perform a brief inspection to make certain the tower is ready for operation.
Give the entire system a visual inspection for loose parts, severed cables, oil leaks, hydraulic leaks or fuel leaks. Repair any problems prior to operating the tower.

Technical Compartment Components

This section covers the procedures for using the components of the technical compartment.



Battery Master Switch: This rotary switch is the main power switch for Sky Watch.



UP/Down Momentary Switch: The operator can control the unit's ascent (up) and descent (down) from the technical compartment. NOTE: the officer compartment door must be closed before the tower will operate.

To raise the officer's compartment: Press and hold the "UP: button until the officer compartment has reached the maximum elevation. The compartment will stop moving when it has reached maximum elevation, the pump will continue to run until the button is released.

To lower the officer's compartment: Press and hold the "DOWN" button until the officer's compartment has reached ground level. The compartment will stop moving when it has reached ground level, release button.



Power Selector Switch: This switch allows the operator to select the "On--Board" generator or "Shore Power" which is an external 120 Volt power source.



110 Volt GFI Outlet: Provides 110 volt power. Generator or Shore power must be selected for this outlet to operate.

Section 6: Operation (cont.)

Officer Compartment

This section describes the function, location and operation of the officer's compartment components. Review and become familiar with the operation of these components. Refer to figure 11 for the location and names of the components.

Officer Compartment Components

This section covers the procedures for using the components. Refer to figure 11 for the location and names of the components.



A/C Controls: Thermostat controls located overhead with easy to read temperature settings.

1. 

Switches #5 & #6: These switches activate the roof mounted spotlights. The number of the switch corresponds to a number on the roof end panels for each light. To activate a light flip the rocker switch up. To turn off the light, flip the rocker switch down.

2. 

Up/ Down Joystick/ Safety button: The safety button **MUST** be depressed while tower is in operation. The joystick controls the ascent and descent of the officer compartment. To elevate the compartment, with safety button depressed lift and hold the joystick until the compartment stops moving, when no movement is felt release the joystick. To descend, with safety button depressed push the joystick down and hold until the compartment stops moving, then release.

3. 

Generator Start/Stop Switch: To start the generator, press and hold the rocker switch in the up position until the generator is running, then release. To shut down the generator, press and hold the rocker switch in the down position until the generator has stopped running then release.

Section 6: Operation (cont.)

4. 

Battery Level Gauge: The unit has two (2) deep cycle batteries that are housed in the technical compartment. This meter indicates the power level that is in the batteries.

5. 
12 Volt DC Outlet: This outlet provides 12 Volts DC @ 10 amps to power hand held spotlights, cellular phones, radios or any accessory that requires 12 volt power.

6. 

Fuel Gauge: This gauge is to keep the operator informed of the fuel level in the fuel tank.

7. 
Tactical / Interior Light Switch: This switch is used to provide a red tactical light for night illumination without giving away your position. The interior side is to give a bright white light illumination.

8. 

Flood Lights (1-4): Each Sky Watch tower is equipped with 4 flood lights located above each window. Below each window there is a number 1 thru 4, these numbers correspond with the light switch number. To turn on each light, lift the corresponding toggle switch to the up position “On”. To turn the light off, press the toggle switch to the down position “Off”.

9. 
110 VAC Outlet (GFI): This outlet is located on the right side of the Control Panel. This outlet is equipped with Ground Fault Interruption. This outlet is designed to automatically open the circuit should it become overloaded or a direct short.

Section 6: Operation (cont.)

10.  **Wind Speed Meter:** This sensor measures the wind speed. Digital model is equipped with an audible alarm with adjustable set points. At 35mph low level alarm will sound. At 45mph a high level alarm will sound alerting the operator to descent as conditions are unsafe.

NOTE: Certain feature components on Sky Watch have their own specific operation instructions need to read thoroughly by all personnel. These include the generator, air conditioner/heater, battery charger, thermal imaging camera system and fire extinguisher.

Section 6: Operation (cont.)

Electrical Options

This section refers to the two (2) 110 Volt AC power sources available. These sources are an on-board generator and shore power (external power). It is essential that the operator be familiar with the electrical considerations covered in this section, for safe and efficient operation of the 110 Volt accessories. Read and understand this section thoroughly before operating.

1. 

On-Board Generator: To operate Sky Watch using the on-board generator, rotate the power selector switch to "Onan Power".

NOTE: Do not operate the Sky Watch tower in an enclosed building while running the on-board generator.

2. **Shore Power:** To operate Sky Watch using Shore Power, rotate the power selector switch (located in the technical compartment see figure 10) to "Shore Power". Using a heavy duty (12 gauge or heavier) grounded extension cord, plug the Sky Watch into a dedicated (single use) circuit outlet rated at 20 amps or better.

NOTE: Shore Power is REQUIRED when operating Sky Watch indoors.

NOTE: Do not attempt to plug too many 110 volt accessories into the outlets onboard. The simultaneous operation of accessories could overload the circuit and cause the generator circuit breaker to trip.

Lift Functions:

NOTE: The "Officer's Compartment" door must be closed and safety button depressed for the Sky Watch tower to elevate from either lift locations.

1. **From Technical Compartment:**
   A. Verify "Master Battery Switch" is in the "ON" position.
   B. To elevate the "Officer's Compartment" depress the "UP" button on the Up/Down Momentary Switch box. Continue to depress button until "Officer's Compartment" is fully elevated. Release button.
   C. To lower the "Officer's Compartment", With safety button depressed push the "Down" button on the Up/Down Momentary Switch box. Continue to depress button until "Officer's Compartment has stopped movement. Release button.

2. **From the Officers Compartment:**
   A. Verify "Master Battery Switch" is in the "ON" position (see figure 10) before entering.
   B. Verify the compartment door is closed and secure:
      1) Open a window prior to closing the door. The compartment has a tight

seal, by opening a window this will allow the air to escape and the door to close and latch easier.

2) Attach the “Door Lanyard”; this lanyard prevents the door from fully opening while in the raised position.

C. To elevate Sky Watch, on the “Control Panel” depress safety button and lift toggle switch and hold until the compartment has reached the top of elevation. Release switch. The tower is now ready for surveillance.

D. To lower Sky Watch, on the “Control Panel” depress safety button and push down and hold toggle switch until the compartment has stopped movement, release switch.

Section 7: Troubleshooting

Troubleshooting Guide

Problem	Probable Cause	Solution
Fails to ascend	Low battery power	Recharge Batteries
	Bad battery connection	Clean and tighten battery connections
	Officer's compartment door is not	Close and secure door
Fails to descend	Low battery power	Recharge Batteries
	Bad battery connections	Clean and tighten battery connections
Moaning and vibration (usually during ascent)	Unit has not been properly lubricated	Refer to periodic maintenance schedule
	Cylinder seal has dried out	Apply penetrating oil where rod moves out of the cylinder.
Electrical systems not operating	Circuit has been overloaded	Reset generator circuit breaker Reset circuit breakers in technical compartment
	Generator is not running	Restart Generator
	Shore Power cord not plugged into an outlet	Plug cord into an outlet Verify outlet circuit not overloaded
Difficulty in towing	Tower not set in tow mode	Verify unit is in tow mode, refer to Section 4, Coupling Unit to Tow Vehicle
	Excess weight in Officers Compartment	Verify extra items are not stowed in Compartment.
	Tires not properly inflated	Verify tire are properly inflated, refer to Section 8, General, Tires for proper inflation schedule

Section 8: Maintenance

General

Establish and adhere to a definite schedule of maintenance and service. If Sky Watch is subjected to extreme operating conditions, consult an ICx New Heights Corporation, Customer Support Specialist, to determine the appropriate maintenance schedule.

Keep an accurate log of all maintenance and service, for warranty support.

Perform all maintenance at the operational hours as indicated. Use this schedule to determine the maintenance required. Refer to the sections following this schedule for proper maintenance procedures.

Service Item	E very Day	First 50 Hours	E very Month	Every 150 Hours	Every 500 Hours	Every 1000 Hours
General Inspection	X [9]					
Check Batteries	X					
Check Engine Oil	X					
Check Engine Coolant	X					
Change Engine Oil and Filter		X [1]		X [2,3,4]		
Test GFI Outlets			X			
Check Tire Inflation				X [8]		
Grease Fittings				X		
Check Wheel Bearings				X		
Replace Engine Air Filter					X [2,4]	
Replace Fuel Filter					X	
Check Engine Coolant Protection					X	
Flush Engine Coolant System						X [5]
Replace Coolant Pressure Cap						X [5]
Replace Engine V-Belt						X [6,7]
Clean Crankcase Breather						X [6,7]
Replace Coolant Hoses and Thermostat						X [6,7]
Replace Batteries						X [6,10]

1. As part of engine break-in, change the engine oil after the first 50 hours of operation.
2. Perform more often when operating in dusty conditions.
3. Perform more often when operating in hot weather.
4. Perform at least once a year.
5. Perform at least once every two years.
6. Perform at least once every five years.
7. Must be performed by a qualified mechanic.
8. Inflate 15” tires to 55 psi and 16” tires to 70 psi.
9. Check for any leaks of hydraulic fluid and make notes of any maintenance concerns.
10. Battery life may be longer or shorter depending the amount of use.

Section 8: Maintenance (cont.)

General Inspection

Perform a general inspection each time before operating Sky Watch. Check the tower for any hydraulic leaks (hoses, fittings and cylinders), broken or loose parts, fuel leaks and any reduction in tire pressure. If there are any issues found, make note of them and have the issue corrected immediately.

Lubrication System

Your Sky Watch tower is equipped with grease fittings at each pivoting joint. These grease fittings are located on top of each outrigger (4) and at the pivot joints of both lifting arms and stabilizer arms (total of 8). These joints are lubricated prior to shipment to you. Proceed with regular lubing as indicated in the maintenance chart.

Battery Care

The control panel, in the Officers Compartment, is equipped with a battery level indicator. This indicates the level of charge the batteries currently hold. The batteries are a vital part of the Sky Watch system, it is important that the batteries maintain a full charge.



NOTE: Sky Watch comes equipped with two charging sources for the 12 volt batteries. The on-board generator supplies power to the battery charger, If Shore Power is used, this source also supplies power to the battery charger.

Hydraulic System

Fill cap is located . Only add fluid if Sky watch does not elevate to full height, contact your Customer Support Center to determine if this is the cause.



NOTE: Use only “Dextron II” transmission fluid and is to be filled to within ¼” of the reservoir top. Only measure the reservoir when the officer’s compartment is on the ground, not on the transport gate. Measuring and filling with the officer’s compartment on the transport gate will overfill the reservoir.

Section 8: Maintenance (cont.)

Generator

Refer to the generator operator's manual for details about maintenance on your generator .Your Sky Watch tower is equipped with an oil drain hose and valve for quick and convenient oil changes.



NOTE: Make sure the oil drain valve is securely closed before refilling with oil.

Obtaining Support or Parts

When your Sky Watch tower requires service, contact ICxNew Heights Corporation, Customer Support Center for assistance. Factory trained representatives are prepared to meet your support needs. Please call 1.800-819-1215 office or visit our 24hour support site at www.customersupport@icxnewheights.com.

When contacting Customer Support, please provide the complete serial number listed on the Data plate that is located on the tongue of the trailer.

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